

A Data-driven Risk Assessment Method for Autonomous Vehicles Without Expert Rule Design

Caojun Wang¹, Shuo Yang¹ and Yanjun Huang¹

Abstract—Autonomous vehicles (AVs) enhance driving efficiency and reduce accidents but require robust risk assessment methods due to dense traffic and uncertainties. Existing methods rely on predefined rules, which lack generalization. This paper presents a novel risk quantification method without expert rules, leveraging reinforcement learning and adversarial agents. The proposed model uses Gated Transformer Networks for multivariate time series regression, analyzing historical traffic data to generate continuous risk assessments. Simulation experiments validate the method's efficacy, demonstrating its precision and robustness.

risk assessment, autonomous vehicles, multivariate time series regression

I. INTRODUCTION

According to NHTSA statistics, approximately 94% of traffic accidents are caused by human drivers [1]. Autonomous vehicles hold promise for replacing humans in driving tasks, potentially reducing accidents. However, the open road is characterized by high density, frequent interactions, and high uncertainty, making AVs vulnerable in real-world situations, as evidenced by Waymo's 19 contact accidents in Phoenix City [2]. This proves that autonomous vehicles at this stage are not 100% safe, and completely avoiding accidents remains a challenge [3].

Risk, closely related to safety, can be defined as the likelihood and severity of potential damage. Risk assessment bridges the gap between perception and planning in autonomous driving systems [4]. Various risk assessment methods have been proposed, including predefined rule-based, probabilistic statistics-based, equivalence model-based, and learning-based methods.

Predefined rule-based methods develop rules and metrics to assess AV risks, considering factors like traffic rules, traffic flow, road conditions, and environmental conditions. Examples include Time Headway (THW) [5] and Time to Collision (TTC) [6], which measure risk based on vehicle distances and speeds.

Probabilistic statistics-based methods use statistical principles to assess risks. Studies have focused on collision probabilities [7] [8], such as using logistic regression to determine risk levels at intersections [9] and applying Extreme Value Theory to evaluate collision probabilities [10].

Equivalence model-based methods assess risks by establishing relationships between system states and risks.

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Artificial potential field theory, initially used in robotics [11], has been adapted for AVs to maintain safe distances from obstacles [12].

However, predefined rule-based method needs a lot of human design, lacking in versatility; probabilistic statistics-based method requires massive amount of data and more complex model; equivalence model-based method is easy to understand, but it highly depends on the model choice and its simplified conditions may also conflict with the real situation. Learning-based method refers to using machine learning algorithms to analyse data, learn patterns and trends from it, and predict potential risk. It gets rid of predefined rules and does not need data as much as probabilistic statistics-based method requires, demonstrating flexibility and advantages to some extent.

There are many existing learning-based approaches in academia. Measured Headway distance and velocity are used to train a model based on BP NN to warn the emergency [13]. Wang et al. use the K-means cluster method to classify near-crashes into different-risk levels based on the three braking process features [14]. Road status acquired from infrastructure is input into Multi-Layer Perception Neural Network to get real-time rear-end collision warning [15]. However, supervised learning or semi-supervised learning inevitably requires the use of labeled data, which is costly to produce or obtain.

To utilize the advantages of learning-based method while eliminating the labeled data, this paper proposes this method with three main contributions:

1. Leveraging the idea of confrontation, a novel definition of risk has been proposed with the help of RL-based adversarial agent;
2. Building on the definition, strategies that lead to the different risk level have been found and a varied risk level dataset has been constructed;
3. Propose a transformer-based risk quantification methodology that does not add any expert knowledge and quantifies the current risk in a multivariate time series regression model.

The overall architecture of risk assessment method is shown in Fig. 1. The paper is organized as follows. The definition of risk is introduced in Section II. The descriptions of varied risk level dataset construction are introduced in Section III. The risk assessment methodology is introduced in Section IV. Section V illustrates the experiment and Section VI concludes the paper.

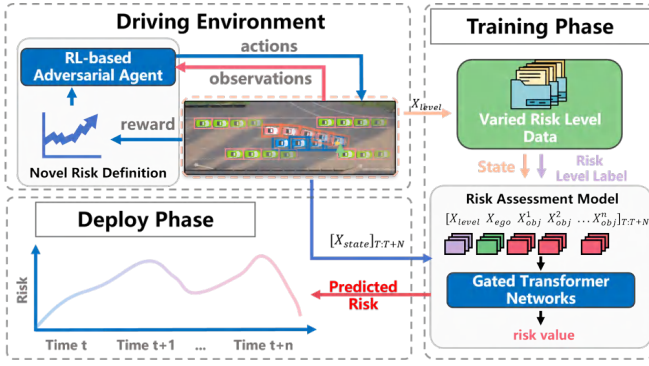


Fig. 1. Overall architecture of risk assessment method for autonomous vehicle

II. DEFINITION OF RISK

A. Problem Statement

Risk is not currently described by a uniform definition and can be expressed in general terms as the degree or the possibility for autonomous vehicles suffering from damage. In this paper, a novel definition of risk is proposed which introduces the idea of confrontation, considering risk as the degree of viciousness to hamper ego vehicle from operating properly. In the driving environment, the more the adversarial vehicle takes actions that prevent ego vehicle from driving normally, the higher the risk is. An explicit definition of risk will be regarded as an important reference for the researchers to identify the direction of improvement.

Also, a technical tool is needed. The concept of an adversarial agent enters the picture which mainly relies on looking for a policy to maximize its influence on the performance of ego vehicle. This could be interpreted into an optimization problem, as shown below:

$$\theta^* = \arg \max_{\theta} [F(\pi_A, \pi_E(\theta), S_c)], \quad (1)$$

where $F(\cdot)$ is the performance evaluation metric, π_A is the ego vehicle policy, π_E is the environment agent policy with parameter θ and S_c is the scenario dynamics. S_c is used to characterize the state transitions of all agents.

The following will explain how to generate an agent to solve this optimization problem and how to measure risk qualitatively with the help of agent under this definition.

B. Adversarial Agent Generation

The dynamic interaction process among the adversarial agent, the driving environment and the ego vehicle is constructed as a Markov Decision Process (MDP), which can be expressed as:

$$M = (S, A, P, \pi, R), \quad (2)$$

where S is state space, A is action space, P is probabilistic model of state transition, π is policy model and R is reward. Markov property is the core nature of MDP which declares the future state of a system is independent of the historical sequence of states and only relies on its current state.

Reinforcement Learning is a potent approach for tackling MDPs and deriving optimal or sub-optimal policies for learning agents [16] [17]. Among the popular reinforcement learning algorithms, the Soft Actor-Critic (SAC) [18] algorithm stands out. It is a model-free, off-policy algorithm based on actor-critic framework. SAC uses stochastic policy, which can make the policy as random as possible. This approach enables agents to explore the state space S comprehensively, mitigating the risk of early convergence to local optima and facilitating exploration of diverse feasible solutions to enhance robustness against interference.

Typically, the ultimate goal of a reinforcement learning algorithm is to find a strategy that maximizes the value of the accumulated rewards, but SAC also seeks to solve the following maximum entropy problem in order to reach the goal of stochastic policy:

$$\pi^* = \arg \max_{\pi} \mathbb{E}_{(s_t, a_t) \sim \rho_{\pi}} \left[\sum_t r(s_t, a_t) + \alpha \mathcal{H}(\pi(\cdot | s_t)) \right], \quad (3)$$

where $\mathcal{H}$ is entropy, α is the temperature parameter. With $\alpha \rightarrow 0$, the maximum entropy RL gradually approaches the conventional RL.

Soft policy iteration alternates between policy evaluation and policy improvement until convergence. The aim of soft policy improvement is to find a superior policy π_{new} , compared to the current policy π_{old} . This is done by representing the policy as a Gaussian distribution and minimizing the Kullback-Leibler divergence to narrow the gap between the current and new policies.

$$\pi_{new} = \arg \min_{\pi' \in \Pi} D_{KL} \left(\pi'(\cdot | s_t) \parallel \frac{\exp(Q^{\pi_{old}}(s_t, \cdot))}{Z^{\pi_{old}}(s_t)} \right), \quad (4)$$

where $Z^{\pi_{old}}(s_t)$ is the normalized distribution of Q values, it will not contribute to the policy gradient, so it can be ignored.

Reward function $r(s_t, a_t)$ is one of the important components of reinforcement learning, which can help reinforcement learning algorithms to study the optimal strategy for the agent. A reasonable reward function design can guide the agent to learn and evolve to get the maximum reward.

Previous study in adversarial agent generation [19] [20] usually sets the reward function opposite from the ego vehicle. However, simply applying this reward function will lead to unreasonable behavior such as stopping in the middle of the lane. Therefore, mechanism rules should be combined in the reward function or control algorithm to constrain the adversarial agent [21]. When combined in the reward function, reward function can be expressed as:

$$r_{adv} = -r_{ego} + \beta r_{rule} \quad (5)$$

where r_{ego} denotes the performance of the ego vehicle, r_{rule} denotes reward originating in the mechanism rules under different cases, preventing the adversarial agent being irrational, and β is the hyperparameter. The concrete state space, action space and reward function will be introduced in experiment phase.

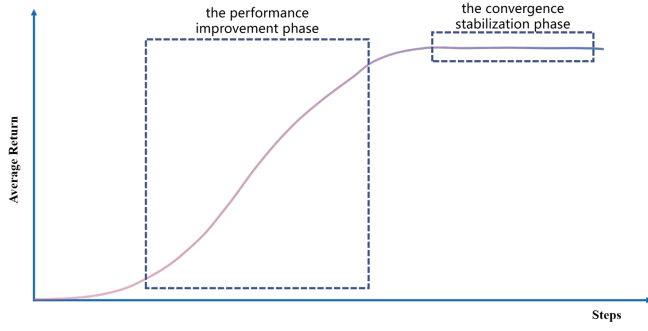


Fig. 2. Typical reward graph for RL

C. Qualitative Measurement of Risk

Definition is not enough. We also need to construct qualitative perceptions of how risk ranges from mild to severe. The process of reinforcement learning from divergence to convergence can be characterised as a gradual stabilisation of the reward function, converging to some interval. During this process, the adversarial agent keeps replacing current policy π_{old} with superior policy π_{new} according to Eq. 4. The process of continuous updating of the policy can be seen as a process of increasing viciousness to the ego vehicle, i.e. increasing risk.

Fig. 2 is a typical reward graph for RL. A reinforcement learning training process with stable convergence can be divided into two phases, i.e., the performance improvement phase and the convergence stabilization phase. In the performance improvement phase, the average return is still continuously increasing, which indicates that the policy search is still ongoing and the model parameters are still being updated to pursue better performance. In the convergence stabilization phase, however, the average return remains basically unchanged, indicating that the policy search is basically over, and the obtained policy is already the optimal policy that the current algorithm can achieve.

Considering that the average return increasing during the performance improvement phase is nonlinear, a qualitative measurement method of risk is proposed to ensure that the distinction is clear and reasonable between the level of risk. The method is shown in Algorithm. 1 and Fig. 3.

The process of measuring risk is shown in Line 6 - Line 19 of Algorithm. 1. First, five different levels of risk are set and the corresponding training steps are obtained through the mapping relation between the average return graph and the training steps, which is defined as:

$$R_s = \Gamma(n_s), \quad (6)$$

where R_s is the corresponding average return at n_s step.

As shown in Fig. 3, the bottom of the horizontal axis represents the collected sample $H_1, H_2, H_3, \dots, H_{\theta_t}$ at each episode during the performance improvement phase. Finally, the samples that fit the corresponding training steps will be placed in risk level group $\mathbb{C}$, which means the qualitative measurement of risk including mild, moderate, severe, very severe and worst is done.

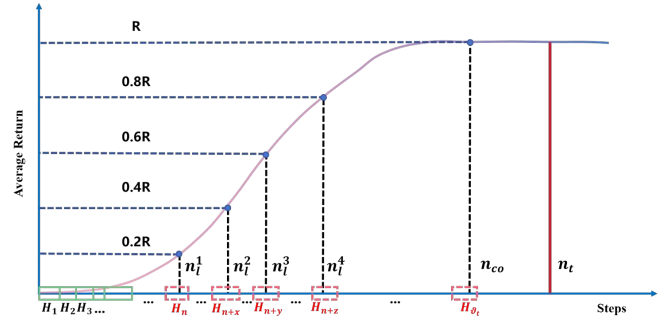


Fig. 3. Qualitative measurement of risk

Algorithm 1 Qualitative Measurement of Risk

- 1: **Initialize:** Risk level group $\mathbb{C} = \emptyset$, set $\Omega = \emptyset$, adversarial agent network π_E with weights θ , step number $n = 0$;
- 2: **Set parameters:** Total training steps n_t ;
- 3: **Input:** Ego vehicle policy π_A , scenario dynamics S_{sd} ;
- 4: **Execute:** Trial train adversarial agent π_E under π_A , n_t and S_{sd}
- 5: **Return:** Total episodes ϑ_t , performance improvement steps n_{co} , maximum average return R
- 6: **for** $i=1, 2, 3, 4, 5$ **do**
- 7: $n_l^i = \Gamma^{-1}(0.2i \cdot R)$
- 8: **for** $i=1, 2, \dots, \vartheta_t$ **do**
- 9: $n_1 = n$
- 10: **Execute** collecting samples, update adversarial agent network π_E using Eq. 3
- 11: $n = n + n_i$
- 12: Collect $H_i = (\pi_i, n_{start} = n_1, n_{end} = n)$
- 13: $\Omega \leftarrow \Omega \cup \{H_i = (\pi_i, n_{start}, n_{end})\}$
- 14: **if** $n > n_{co}$ **then**
- 15: **break**
- 16: **for** $i=1, 2, 3, 4, 5$ **do**
- 17: **for** $j=1, 2, \dots, \vartheta_t$ **do**
- 18: **if** $n_l^i > \Omega[j][1]$ and $n_l^i < \Omega[j][2]$
- 19: $\mathbb{C} \leftarrow \mathbb{C} \cup \Omega[j]$

III. VARIED RISK LEVEL DATASET CONSTRUCTION

The risk level group $\mathbb{C}$ can be used to build a varied risk level dataset and this process is carried out in a simulation environment. The parameters of the simulation environment are randomly configured, which is to simulate more diverse real-world inputs and ensure the generalization of subsequent model training.

The procedure of the varied risk level dataset construction is shown in Table 2.

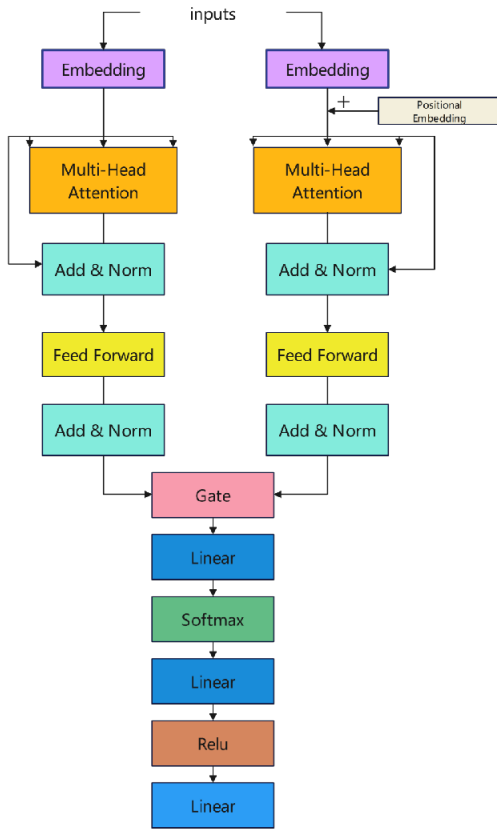


Fig. 4. Structure of modified Gated Transformer Networks

Algorithm 2 Varied Risk Level Dataset Construction

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1: Initialize: Simulation environment  $S_{world}$ , varied risk
   level dataset  $\Psi$ , time step  $t$ ;
2: Input: Risk level group  $\mathbb{C}$ , traffic model  $\zeta_{tf}$  with pa-
   rameters  $\sigma$ ;
3: Set parameters: Same level data size  $n_{sc}$ , episode
   number  $n_{episode}$ ;
4: for  $i=1, 2, 3, 4, 5$  do
5:   for  $j=1, 2, \dots, n_{episode}$  do
6:     Set  $t=0$ 
7:     Deploy the policy  $\mathbb{C}[i]$  to the adversarial agent
8:     Random choose traffic model parameters  $\sigma$ 
9:     Generate random traffic flow using  $\zeta_{tf}$  in simu-
       lation
10:    Collect  $M_j = (t, S, 0.2 * i)$ 
11:     $\Psi \leftarrow \Psi \cup M_j = (t, S, 0.2 * i)$ 
12:    if total time steps  $> i * n_{sc}$  then
13:      break

```

Line 4 - Line 13 of Algorithm. 2 introduce how to construct varied risk level dataset. In Line 10 - Line 11 of Algorithm. 2, the time steps, the state and the risk level is collected to varied risk level dataset Ψ , which can be denoted as:

$$\Psi = \{M_1, M_2, M_3, \dots\} \quad (7)$$

where $M_j = (t, S, 0.2 * i)$, j represents the episode number,

$0.2 * i$ denotes the five different risk level from 0.2 to 1.

The algorithm proceeds to exit the current inner loop, moves on to implement the policy for the next risk level, and repeats this process until varied risk level datasets Ψ are fully collected.

IV. RISK ASSESSMENT METHODOLOGY

Varied risk level dataset consists of the time-dependent state. The collected state can be thought of as multivariate time series, a collection of data in which multiple variables are observed and recorded at multiple points in time, which have richer information and more complex structures than traditional one-dimensional time series, and allow for the simultaneous observation and analysis of the relationships and trends between multiple variables.

Multivariate time series classification has been extensively studied by the academic and has a wide application in domains like healthcare, finance, industrial engineering and IoT [22]. Deep learning models for multivariate time series classification include Convolutional Neural Networks [23], LSTMs [24], Transformer [25] and so on.

In risk assessment, however, the classification only gives qualitative risk levels, i.e. Mild, Moderate, Severe, Very Severe, Worst, but it does not allow for a more nuanced differentiation between the levels to be explored. Continuous quantitative values can contain more information and better mine the data for subtle changes and trends, helping people to better understand changes in risk.

In this paper, we use multivariate time series regression based on the Transformer without any expert rule design to assess risk. The following will explain the Transformer networks we use and the process of training a risk assessment model.

A. Modified Gated Transformer Networks

The proposed risk assessment methodology is constructed based on Gated Transformer Networks (GTN) [26]. It is an extension of current Transformer networks, which is naturally and effectively suitable for the multivariate time series classification task. The structure is shown in Fig. 4.

Its specialty lies in two-tower transformer with a gating. Two towers are responsible for capturing the step-wise (temporal) and channel-wise (spatial) feature respectively, while the gating merges two towers of transformer networks. This enables GTN explicitly learn both channel-wise and step-wise correlations and achieve competing performance with current state-of-the-art deep learning models in Multivariate Time Series Classification.

The output layer has been modified that we add a linear layer, a Relu activation layer and another linear layer in sequence to the original networks in order to make output a continuous value.

B. Training

The input of the Modified Gated Transformer networks is a Multivariate Time Series plus its corresponding risk label X_{level} , which is a collection of the relative information

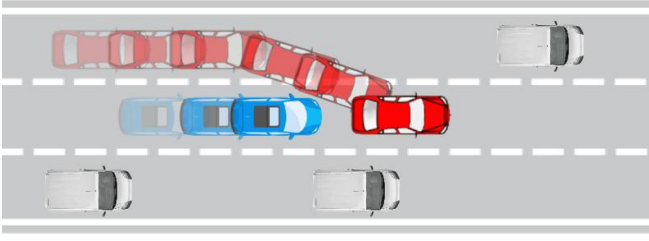


Fig. 5. Cut-in scenario

between the adversarial agent and the ego vehicle for a specific time period. The input vector can be expressed as follows.

$$\mathbf{X}_{T:T+N} = [X_{level}, S]_{T:T+N} \quad (8)$$

The training process of risk assessment model κ is shown in Table 3. κ is trained with data from varied risk level dataset Ψ .

Algorithm 3 Training process of risk assessment model κ

Initialize: Risk assessment model κ with weights θ_{risk} , multivariate time series dataset ϕ ;
Input: Varied risk level dataset Ψ ;
Set parameters: Batch size n_b , epoch number n_{epoch} , time step length N ;
for $i=1, 2, \dots, \text{len}(\mathbf{C})$ **do**
 for $j=1, 2, \dots, \text{len}(\Psi)$ **do**
 for $T=0, 1, 2, \dots, \text{max}(\Psi[j][1]) - N$ **do**
 Sample $[X_{level}, S]_{T:T+N}$ from $\Psi[j]$
 $\phi \leftarrow \phi \cup [X_{level}, S]_{T:T+N}$
 for $i=1, 2, \dots, n_{epoch}$ **do**
 Random sample $B = \{[X_{level}, S]_{T:T+N}\}_{k=0:n_b-1}$
 from ϕ
 Predict $\hat{a} = \kappa_{\theta_{risk}}[X_{level}, S]_{T:T+N}$
 Compute MSE loss $\mathcal{L} = \|a - \hat{a}\|^2$
 Update θ_{risk} by $\hat{\nabla}_{\theta_{risk}} \mathcal{L}_{\kappa}(\theta_{risk})$ using AdaGrad optimizer
 end for

V. EXPERIMENT

In this section, we demonstrate the feasibility and advantages of the proposed methods through experiments. These experiments are validated in simulation platform CARLA [27]. Testing all these methods need a specific scenario and the cut-in scenario is chosen because the most common behaviour that may affect the operation of an autonomous vehicle is cut-in.

A. Settings

As shown in Fig. 5, illustrating how the adversarial agent conducts cut-in behavior to hamper the ego vehicle from operating. The blue car represents ego vehicle, the red one represents adversarial agent and the white ones represent other traffic vehicles. The adversarial agent will drive in the neighbouring lane of the ego vehicle, and after completing

the overtaking manoeuvre, will complete the cut-in at a suitable distance and influence the ego vehicle as much as possible by manipulating the throttle, brake, and steering to cause a collision.

For this scenario, state space S , action space A and reward function R need to be declared according to Eq 2.

State space needs to let agent get enough information about the environment, which is defined as follows:

$$S = [\Delta s \quad \Delta d \quad \Delta v_s \quad v_s \quad a_s], \quad (9)$$

where Δs is the relative longitudinal distance between the environment agent and the ego vehicle, Δd is the relative lateral distance, Δv_s the relative longitudinal velocity, v_s and a are the longitudinal velocity and the longitudinal acceleration of the environment agent, respectively.

In the experiment, the quintic and quartic polynomial curves are used in the lateral and longitudinal trajectory planning process of adversarial agent. Considering the boundary condition from polynomial curves, the action space is defined as:

$$A = [d_{fn} \quad \dot{s}_f], \quad (10)$$

where $\dot{s}_f$ is the expected longitudinal velocity after the planning time T_c . The PID controller is applied to tracking the desired trajectory generated by the reinforcement learning algorithm.

Reward function is designed as follows, considering both mechanism rules and adversarial behavior.

The risk reward r_1 is designed for the sake of reflecting adversarial policy that guides the adversarial agent to hamper the ego vehicle from driving normally. Inspired by the improved artificial potential field [28], r_1 is expressed as:

$$r_1 = \sum_{i=1}^m \min \left(\exp \left(\frac{(\min(\max(\Delta d - L_a, r_{\min}), r_{\max}))^2}{\delta_1^2} + \frac{(\min(\max(\Delta s - L_b, r_{\min}), r_{\max}))^2}{\delta_2^2} \right), k_f \cdot g + b, 0 \right) \quad (11)$$

where $r_{\min}$ and $r_{\max}$ are the lower safety distance penalty coefficient and upper safety distance penalty coefficient, respectively. δ_1 and δ_2 are lateral repulsive force influence factor and the longitudinal repulsive force influence factor, respectively. L_a and L_b are the width and length of ego vehicle, respectively. k_f is the scale factor. In order to keep the reward values within a reasonable range, define g as the linear mapping proportionality quantity and b as the linear mapping deviation quantity.

Speed is crucial for adversarial agents, and a reasonable speed-dependent reward function can guide the agent to accelerate from 0 in the starting phase and maintain it in a suitable speed interval, preventing agent from being irrational. The speed reward r_2 is designed as follows:

$$r_2 = -\rho_L \times \frac{v_s}{v_{\min}} - \rho_H \times \frac{v_s - v_{\min}}{v_{\max} - v_{\min}}, \quad (12)$$

where ρ_L is the start-up stage reward factor, ρ_H is the speed maintenance stage reward factor, v_s is the vehicle speed,

TABLE I
PARAMETERS SETTING

Parameters terms	Symbol & Value
Lower safety distance penalty coefficient	$r_{\min} = 0.0$
Upper safety distance penalty coefficient	$r_{\max} = 150.0$
Lateral repulsive force influence factor	$\delta_1 = 8.0$
Longitudinal repulsive force influence factor	$\delta_2 = 10.0$
Scale factor	$k_f = 0.001$
Vehicle width	$L_a = 2.077m$
Vehicle length	$L_b = 5.037m$
Linear mapping proportionality quantity	$g = 2086$
Linear mapping deviation quantity	$b = -20$
Start-up stage reward factor	$\rho_L = 0.5$
Speed maintenance stage reward factor	$\rho_H = 4.0$
Expected minimum speed	$v_{\min} = 7.5m/s$
Expected maximum speed	$v_{\max} = 22.0m/s$
Collision reward factor	$\rho_{coll} = 200.0$
Simulation step size	$\Delta t = 0.1s$
Discount factor of value network	$\gamma = 0.99$
Learning rate of adjust temperature	$\lambda = 3e - 4$
Learning rate of target network	$\tau_1 = 5e - 3$
Total training steps	$n_t = 100000$
Same level data size	$n_{sc} = 10000$
Episode number	$n_{episode} = 1000$
Batch size	$n_b = 32$
Epoch number	$n_{epoch} = 1000$
Time step length	$N = 40$
Lane Width	$L_{lane} = 3.5m$

$v_{\min}$ and $v_{\max}$ are the desired minimum and maximum speed, respectively.

Collision is considered to be the most representative result that would result from adopting adversarial behaviour. Similarly, the adversarial agent is encouraged to cause collision to force the ego vehicle to be unable to continue its operation. Therefore, the collision reward is designed as shown below:

$$r_3 = \begin{cases} 0 & \text{not collision} \\ q_{coll} & \text{collision} \end{cases}, \quad (13)$$

where q_{coll} is collision reward factor that should be set to a positive value, meaning collision is encouraged.

The total reward function is:

$$R = -r_1 + r_2 + r_3 \quad (14)$$

The parameters of reward function and the settings of experiment is shown in Table I.

B. Evaluation

We designed a training environment based on highway in Town 4 in CARLA. To generate random traffic flow, the spawn point and desired speed of other vehicles are randomly selected. After 100,000 training steps, the learning curve of average return is shown in Fig.6. According to algorithm 1, five policies representing different levels of risk are extracted and deposited into the risk level group C. The varied risk level dataset is constructed by loading different policies into the adversarial agent in the environment. We use t-SNE to evaluate the proposed qualitative measurement of risk. T-SNE(t-distributed Stochastic Neighbor Embedding) is a non-linear dimensionality reduction technique used to map high-dimensional data into 2D or 3D space for visualisation or further analysis [29]. T-SNE is mainly used in clustering

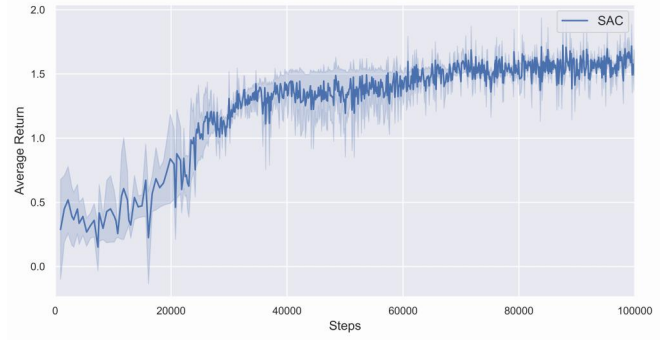


Fig. 6. Learning curve of the average return of SAC

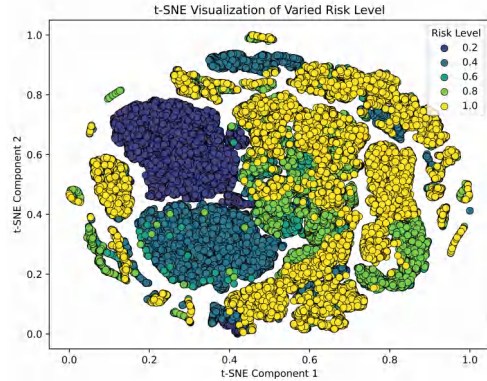


Fig. 7. T-SNE visualization of varied risk level

or classification tasks. In this paper, the t-SNE method is applied to process the data in the varied risk level dataset. From Fig. 7, it can be seen that the data of different risk levels has obvious aggregation which proved that the proposed method can capture different levels of risk.

Afterwards, the risk assessment model is obtained according to algorithm 3. In order to evaluate its effectiveness and robustness, a new adversarial vehicle has been introduced in CARLA by setting different velocity and cut-in distance.

Various velocity: As shown in Fig. 8, adversarial vehicle's velocity is set in three different mode: constant, phased change and sharp change while cut-in distance remains 6m. The value of risk is around 0.4 at the beginning and meets a surge when the adversarial vehicle approaches the ego vehicle and starts to overtake. Fig. 8(a) shows that adversarial vehicle completes cut-in maneuver and keeps driving at around 14.5m/s while risk lowered to near 0.2. Fig. 8(b) shows the adversarial vehicle maintains different speeds at different stages of time: after completing cut-in, it maintains 16m/s; after moving away, it gradually reduces its speed and shortens the relative longitudinal distance, with a corresponding rise in risk. Fig. 8(c) shows risk stays at a high value when the adversarial vehicle conducts rapid deceleration after cut-in, and then the collision happens.

Various cut-in distance: As shown in Fig. 9, adversarial vehicle's cut-in distance is set to 5m, 7.5m and 10m while velocity remains around 13.5m/s. The value of risk meets a surge again when the adversarial vehicle starts to cut in

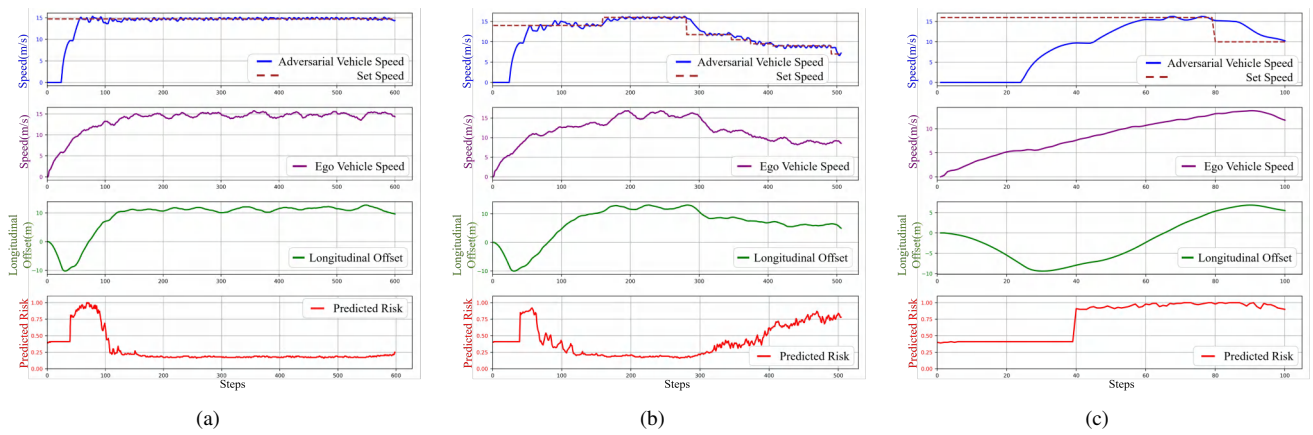


Fig. 8. The velocities, longitudinal offset and predicted risk between the ego vehicle and the adversarial vehicle under various velocity mode. (a) Constant. (b) Phased change. (c) Sharp change.

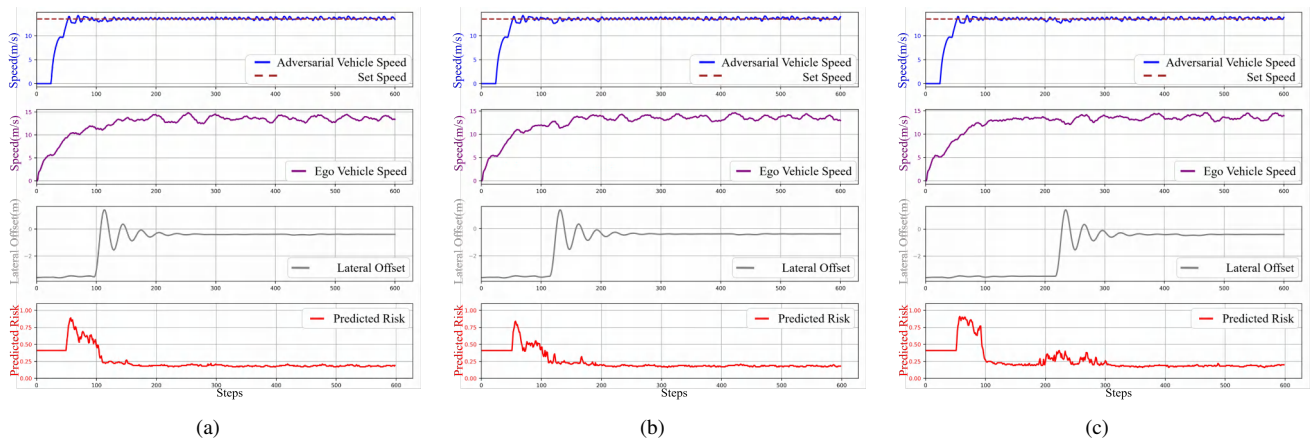


Fig. 9. The velocities, lateral offset and predicted risk between the ego vehicle and the adversarial vehicle under various cut-in distance. (a) 5m. (b) 7.5m. (c) 10m.

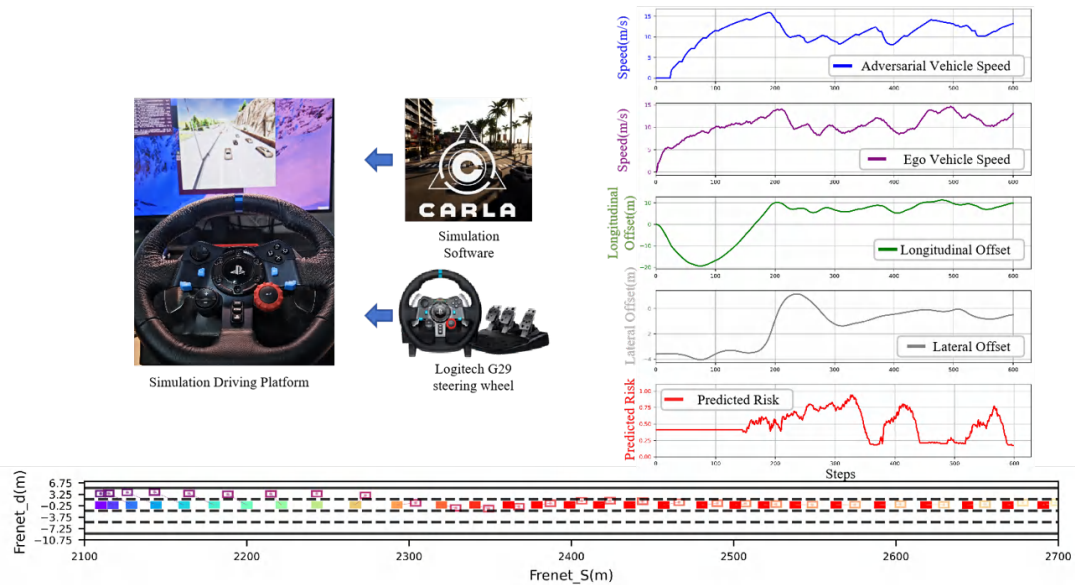


Fig. 10. The trajectory, velocities, longitudinal offset, lateral offset and predicted risk between the ego vehicle and the adversarial vehicle driven by human. (Hollow box represents adversarial vehicle and the solid one represents ego vehicle)

as the relative lateral offset is changing and shows different tendencies under various cut-in distance. The mean value of surge shows a decreasing trend as the cut-off distance gradually increases.

However, simply setting the conditions is not enough. In reality, the behavior of traffic vehicles tends to be more unpredictable and random. Therefore, an adversarial vehicle is driven by a human driver using the Logitech G29 steering wheel in CARLA to perform cut-in maneuvers.

The adversarial vehicle driven by human: The adversarial vehicle in Carla is controlled artificially through the steering wheel. As shown in Fig. 10, human drivers do not have a pre-set speed or cut-in distance, and their behaviour exhibits a high degree of randomness and uncertainty. Under this situation, the model generated by the proposed method demonstrated robustness to synthesise information and assess risk.

VI. CONCLUSION

In this paper, we proposed a data-driven risk assessment method for autonomous vehicles without expert rule design. The method creates a risk assessment model based on transformer networks with the help of two key components which are the novel definition of risk and the construction of varied risk level dataset. We designed three sets of experiments covering speed variations, cut-in distance variations, and human driver interventions, covering artificially set mechanism conditions as well as stochastic behavioural patterns. According to the results, the proposed qualitative measurement of risk can capture different levels of risk and the model generated by the proposed algorithm can output continuous risk assessment values. In addition, the model shows a certain level of robustness since it can assess the risk under more random and unpredictable behaviour caused by human drivers. Future research will focus on how to apply this method under more sophisticated traffic flows to assess the risk more precisely.

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